HAYSEY'S

ASTROSTACKER

USER MANUAL

Version 1.0.0 · Beta Bronze

Full calibration pipeline · 9 stacking methods

PSF frame rejection · Drizzle super-resolution

Star reduction · Colour balance · Crop tool

Plate solving & WCS embedding · Mosaic builder

Non-Local Means denoise · PSF-informed sharpening

github.com/haysey/astrostacker

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Not for public release
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1. INTRODUCTION

Haysey's Astrostacker is a free, open-source astrophotography image stacking application. It is designed to be approachable for complete beginners while offering the full suite of processing tools that experienced imagers expect.

The core purpose of the app is image stacking: taking many individual exposures (called "frames") of the same object and combining them into one clean, detailed image. Stacking dramatically reduces noise, reveals faint detail, and eliminates transient artefacts such as satellite trails, cosmic rays, and hot pixels.

Key features:

- Full calibration pipeline (darks, flats, dark flats, master frames)
- 9 stacking methods with real-time parameter control
- Automatic frame quality scoring and rejection
- Colour camera debayering (RGGB, BGGR, GBRG, GRBG)
- Star alignment across all frames
- Light pollution gradient removal
- Non-Local Means denoising
- PSF-informed sharpening (Richardson-Lucy / unsharp-mask hybrid)
- Drizzle 2x super-resolution stacking
- Interactive post-processing window (star reduction, colour balance, crop)
- Plate solving via nova.astrometry.net with WCS embedding
- Mosaic stitching from multiple plate-solved panels
- Blink comparator for visual frame inspection
- Session save/load
- FITS, TIFF, JPEG, and PNG export

2. SYSTEM REQUIREMENTS & SUPPORTED FILE FORMATS

Supported operating systems:

- macOS 12 or later (Apple Silicon natively compiled, Intel supported)
- Windows 10 / 11 (x64)
- Linux (x64 AppImage, tested on Ubuntu 22.04+)

No installation required — the app is a self-contained bundle.

Supported input file formats:

- FITS (.fits, .fit, .fts) — the standard format for astronomy cameras
- XISF (.xisf) — used by Sequence Generator Pro, N.I.N.A., etc.

Export formats:

- FITS (linear, full floating-point precision — preferred for further work)
- TIFF (auto-stretched 8-bit, for sharing and further processing)
- JPEG (auto-stretched 8-bit, compressed — for web sharing and posting)
- PNG (auto-stretched 8-bit, for quick previews and sharing)

The app does not currently support raw camera formats (.CR2, .NEF, .ARW, etc.) directly. Use a tool such as AstroPixelProcessor, Siril, or even GIMP/darktable to convert raw files to FITS first, or shoot in FITS mode if your capture software supports it (e.g. N.I.N.A., SharpCap, FireCapture).

Internet connection:

- Required only for the optional plate solving feature.
- All stacking, calibration, and processing runs entirely offline.

3. INTERFACE OVERVIEW

The main window is divided into three regions:

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■  MENU BAR  (File  |  Tools)                                          ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                  ■                                                    ■
■  LEFT SIDEBAR    ■          PREVIEW PANEL                            ■
■  (Tabbed)        ■    (image display + histogram on right)           ■
■                  ■                                                    ■
■  · Stacking      ■                                                    ■
■  · Plate Solve   ■                                                    ■
■  · Mosaic        ■                                                    ■
■                  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                  ■  SETTINGS PANEL      ■  PROGRESS / LOG PANEL       ■
■                  ■  (camera, method,    ■  (Start button, progress    ■
■                  ■   processing opts)   ■   bar, activity log)        ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■  FRAME STATUS BAR  (frame counts, accepted/rejected, version)        ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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Left Sidebar tabs:

- Stacking — Load your light, dark, flat, and dark flat frames here.
- Plate Solve — Set up your API key and scale hints; solve images.
- Mosaic — Stitch multiple plate-solved FITS panels into one image.

The splitter handles between panels can be dragged to resize any region.

4. FIRST-RUN SETUP WIZARD

On first launch, a four-page Setup Wizard appears automatically to help you configure the most important settings. You can re-open it any time via:

Tools > Setup Wizard...

Page 1 — Welcome

Explains what image stacking is and what the wizard will cover.

Page 2 — Your Camera

Choose Colour camera or Monochrome camera.

• Colour (Bayer) — Canon, Nikon, Sony DSLR/mirrorless, ZWO ASI colour,

```
    Altair colour, Player One colour, and any other one-shot-colour camera.
```

• Monochrome — dedicated black-and-white astronomy cameras, typically used

```
    with separate colour filters (LRGB, Ha, OIII, SII, etc.).
```

If you are unsure: if your images come out in colour straight from the camera or capture software, choose Colour.

Page 3 — Plate Solving (Optional)

Enter three values that help the plate solver find your image quickly:

• API Key — a free key from nova.astrometry.net (see Section 10.2)

• Focal length (mm) — your telescope's focal length

• Pixel size (µm) — your camera's pixel size in micrometres

You can skip this page and set it up later in the Plate Solve tab. See Sections 11.4 and 11.5 for how to find these values.

Page 4 — You're Ready!

A quick-start guide showing you the five steps to your first stack.

Everything entered in the wizard is saved automatically. Your settings persist across app launches using your operating system's settings store.

5. THE STACKING TAB — LOADING YOUR FRAMES

The Stacking tab in the left sidebar contains four frame lists. Only Light Frames are required — all other frame types are optional but improve results.

5.1 LIGHT FRAMES

Light frames are your actual exposures of the sky — the science images you took through the telescope. These are the only mandatory input.

To add light frames, click the "Add" button and choose either:

- Add Files... — opens a file picker for individual files
- Add Folder... — scans a folder and adds all supported files from it

The frame count is shown in the section header (e.g. "LIGHT FRAMES (47)").

Clicking a file in the list shows a preview of it in the Preview panel and sets it as the target image for a manual plate solve.

5.2 DARK FRAMES

Dark frames are exposures taken with the lens cap on (or camera covered) at the same temperature, gain, and exposure duration as your light frames. They capture the thermal noise (hot pixels, amp glow, dark current) generated by the sensor itself, which are then subtracted from each light frame.

Best practice:

- Take at least 10–20 darks per imaging session.
- Match exposure time exactly to your light frames.
- Match camera temperature (critical for cooled astronomy cameras).
- You do not need to retake darks if temperature and settings match a

```
    previous session — you can use a saved master dark instead.
```

If you omit dark frames, the pipeline will skip dark subtraction. This is often acceptable for cooled cameras at low temperatures, or very short exposures where dark current is negligible.

5.3 FLAT FRAMES

Flat frames are evenly-lit exposures of a uniform light source — typically a flat panel, dawn/dusk sky, or white T-shirt over the front of the telescope.

They capture vignetting (darkening toward the corners caused by the optics) and dust motes on the sensor or filters, which are then divided out from each light frame.

Best practice:

- Take at least 10–20 flats per setup.
- Keep the focus position identical to your light frames.
- Flats must be taken with the same filter/camera rotation as your lights.
- Retake flats if you move the camera, rotate a filter, or refocus.
- Aim for a histogram midpoint around 30–50% of full well capacity.

5.4 DARK FLAT FRAMES

Dark flat frames (also called "flat darks" or "bias frames") are short exposures taken with the camera covered, at the same exposure duration as your flat frames. They are used to calibrate the flats before the flats are used to calibrate the lights — a full calibration chain.

For most cooled cameras and modern DSLRs, bias frames or dark flats are not strictly necessary, but they improve accuracy, especially if your flat exposure is more than a second or two.

5.5 MASTER DARK / MASTER FLAT (PRE-BUILT)

If you already have a master dark or master flat from a previous session (or built in another application), you can load them directly using the "Master Dark..." and "Master Flat..." buttons at the bottom of the Stacking tab. Loading a master frame overrides any individual frames in the list above.

After loading, the filename is shown below the button as confirmation.

The pipeline also saves the master dark and master flat it builds to the same folder as your stacked output, named "master_dark.fits" and "master_flat.fits". You can reload these in future sessions to save time.

5.6 DRAG-AND-DROP & FOLDER IMPORT

All four frame lists support drag-and-drop from your file manager or desktop. You can drag individual files or entire folders onto a list — folders are automatically scanned for supported image formats.

The "Add Folder..." option in the Add button menu does the same thing via a folder picker dialog.

Duplicate files are automatically skipped when adding to a list.

6. SETTINGS PANEL — CAMERA & PROCESSING

The Settings panel sits at the bottom of the main window. Use the scroll bar inside it to see all options.

6.1 CAMERA TYPE

Mono — Monochrome camera. No debayering is performed.

Colour (Bayer) — Colour / one-shot-colour camera. The app will debayer

```
                    each frame from its raw Bayer mosaic to RGB before stacking.
```

This must match your actual camera. If set to Mono with a colour camera (or vice versa), the stack will not turn out correctly.

6.2 BAYER PATTERN

Only visible when Camera Type is set to Colour (Bayer).

The Bayer pattern describes the arrangement of red, green, and blue photosites on your sensor. It is a fixed property of the camera model and does not change between sessions.

RGGB — Most common. Default for ZWO ASI, Altair, Player One, and most

```
            DSLR/mirrorless cameras (Canon, Nikon, Sony).
```

BGGR — Some Sony sensors. Check if colours look wrong with RGGB.

GBRG — Less common. Try if RGGB and BGGR both produce wrong colours.

GRBG — Less common.

If you are unsure, start with RGGB. If your stacked image has swapped colours (e.g. nebula emission regions that should be red appear blue), try BGGR next.

Your camera's specification sheet or manufacturer's website will list the correct value.

6.3 STACKING METHOD

The stacking method determines how pixel values from multiple aligned frames are combined into one final value at each pixel position. The default is Median, which is safe for any frame count and any target.

A full guide to choosing the right method is in Section 14.1. Brief summaries:

Mean (Average)

Takes the mathematical average of all pixel values at each position. Maximum signal-to-noise ratio improvement with clean data. No outlier rejection — a single satellite trail will appear in the result.

Use when: you have very clean data and no hot pixels or trails.

Median

Takes the middle value when all frame values are sorted at each position. Excellent outlier rejection — satellite trails, hot pixels, and cosmic rays are rejected without any parameter tuning.

Use when: any number of frames, any target, especially for first attempts.

Sigma Clipping

Iteratively identifies and rejects pixel values that are more than a set number of standard deviations (sigma) from the median, then averages the survivors. Better SNR than pure Median when you have 15+ frames.

Use when: 15 or more frames, especially when satellite trails are present.

Winsorized Sigma

Like Sigma Clipping but instead of discarding rejected pixels entirely, it clamps them to the rejection threshold. Preserves more signal, making it better than standard sigma clipping for smaller frame counts (8–15).

Use when: 8–15 frames, similar targets and exposure conditions to sigma.

Percentile Clipping

Sorts all values at each pixel position, discards the bottom X% and top Y%, then averages the rest. Simple and effective with any frame count. Very good at removing satellite trails and planes.

Use when: any frame count, especially with persistent interference.

Weighted Mean (Quality)

Like Mean but each frame is weighted by its sharpness and roundness. Sharper frames with rounder stars contribute more to the result. Measures star PSF profiles automatically before stacking.

Use when: your frame quality varies significantly (e.g. mixed seeing conditions over a long session). Requires at least 3 frames.

Noise-Weighted Mean

```
    Estimates the background noise in each frame using Median Absolute
```

```
Deviation, then weights frames inversely to their noise level. Cleaner
frames contribute more; frames taken through cloud are downweighted.
Use when: some frames were taken through thin cloud or have varying noise.
```

Minimum

Returns the lowest pixel value across all frames at each position.

Diagnostic use — useful for mapping gradients or finding the darkest sky.

Maximum

Returns the highest pixel value across all frames at each position.

Can be used to identify transient objects (satellites, meteors) or for creating star trail images.

6.4 STACKING METHOD PARAMETERS

Some methods expose additional parameters:

Sigma Low / Sigma High (for Sigma Clipping and Winsorized Sigma)

```
How many standard deviations below (Low) and above (High) the median
a pixel must be before it is clipped or clamped. Defaults of 2.5/2.5
work well for most datasets. Lower = more aggressive rejection (more
pixels removed); higher = more permissive (fewer removed).
Typical range: 2.0–3.5
```

Reject Low % / Reject High % (for Percentile Clipping)

```
What percentage of the lowest and highest values at each pixel position
to discard. Default 10%/10% is a good starting point. Increase for
more aggressive rejection of bad frames or satellite trails.
Valid range: 0–49% per side (combined must be less than 100%).
```

6.5 REFERENCE FRAME

The frame that all other frames are aligned to. Default is 0 (the first frame loaded). The spinner's maximum is automatically updated to match the number of light frames loaded.

For best results, choose a frame that is sharp, well-exposed, and has good star coverage. If auto-reject is enabled, rejected frames are removed before alignment so the reference frame index may effectively shift.

6.6 OUTPUT FILE

The file path where the stacked result is saved. Defaults to "stacked.fits" which, inside the app bundle, resolves to the same directory as your first light frame. Click Browse... to choose a specific location and filename.

Supported output formats:

- .fits / .fit / .fts — FITS format (recommended for further processing)
- .xisf — XISF format (compatible with PixInsight)

7. PROCESSING OPTIONS

All processing options in the Processing section of the Settings panel are optional and can be combined freely.

7.1 AUTO-REJECT BLURRY/TRAILED FRAMES

When enabled, the pipeline measures the quality of each calibrated frame before stacking by fitting Gaussian profiles to detected stars. It scores each frame on:

- FWHM (Full Width at Half Maximum): lower = sharper stars
- Eccentricity: lower = rounder stars, higher = trailed or elongated
- Star count: more detected stars generally means better SNR

Frames that score significantly worse than the group median are automatically rejected and excluded from the stack. The progress log shows each frame's scores and whether it was kept or rejected.

Requirements: at least 3 light frames (too few frames for meaningful scoring). If rejection would leave fewer than 2 frames, all frames are kept instead.

7.2 REMOVE LIGHT POLLUTION GRADIENT

Light pollution and vignetting produce a gradual brightness variation across the image background — brighter near the horizon or in the direction of city lights. This option fits a smooth polynomial surface to the background of the final stacked result, then subtracts it, leaving a flat, even background.

Recommended for: suburban or urban imaging sites, moonlit sessions, or any target where the background is noticeably brighter on one side.

Note: Gradient removal operates on the final stacked image, after Auto-crop has removed alignment borders. This means the background sampler has clean data all the way to the image corners — giving accurate gradient modelling even for wide-field targets.

7.3 LOCAL NORMALISATION (PER-FRAME)

Applies the same gradient removal algorithm (see 7.2) to every individual frame before alignment, rather than just the final stack.

When to use it:

Long multi-hour imaging sessions where sky brightness, cloud, or local horizon interference changed significantly between frames. Correcting each frame for its own gradient before combining ensures that varying sky levels do not contaminate the stack.

When NOT to use it:

• For most targets, post-stack gradient removal (section 7.2) gives

```
a cleaner, smoother result and should be preferred.
```

• On targets that fill the entire field of view (large emission nebulae

```
such as the Eta Carinae or Orion nebula complex), per-frame correction
```

```
can produce a mottled, blotchy background in the stack. The background
```

```
sampler has difficulty distinguishing faint nebulosity from sky in every
```

```
individual frame, and the slightly different corrections bake into the
```

```
stack as large-scale texture.
```

• Do not enable both Local Normalisation and Remove Light Pollution

```
Gradient at the same time. Running both creates a double-correction
```

```
that amplifies the mottled effect. Use one or the other.
```

Note: Local Normalisation is slower than post-stack gradient removal because it processes every frame individually.

7.4 AUTO-CROP STACKING EDGES

Frame alignment introduces black or NaN-filled borders around the edges of the stacked result because different frames cover slightly different regions of sky. These borders can cause problems in downstream tools.

When enabled, the pipeline automatically crops the stack to the largest rectangular region fully covered by all frames, then adds a small 2-pixel safety margin. The crop dimensions are reported in the progress log.

Recommended: on, for most sessions with any tracking error or guiding drift.

7.5 DENOISE

Applies Non-Local Means (NLM) denoising to the final stacked image. NLM is a patch-based algorithm that finds similar regions elsewhere in the image and uses them to reduce noise. It is very effective at smoothing grainy background areas while preserving star profiles and fine nebula structure.

No model files or GPU are required — it runs entirely on the CPU.

Strength options:

Light — Subtle smoothing. Safest choice; ideal for high SNR stacks.

Medium — Good balance of noise reduction and detail preservation.

```
Recommended default if you enable this option.
```

Strong — Aggressive smoothing. Best for very noisy short-exposure stacks

```
or datasets with only a handful of frames.
```

Tip: Denoise is most effective when applied after stacking many frames. A stack of 5 frames with Strong denoise will not look as good as a stack of 50 frames with no denoise. Stack more frames first if possible.

Target type matters:

Compact targets (galaxies, globular clusters, planetary nebulae) with clear sky background surrounding the object respond very well to NLM denoising — the algorithm readily finds similar sky patches to average.

Large emission nebulae that fill the entire field of view may look worse with denoising enabled. NLM finds "similar" patches across different parts of the nebula and averages them together, smearing real fine structure and creating unwanted texture. For these targets, try the stack without denoising first — a well-exposed multi-frame stack often needs none.

7.6 SHARPEN

Applies PSF-informed sharpening to the final stacked image. The app first measures the average star Full Width at Half Maximum (FWHM) of your aligned frames using Gaussian fitting, then applies an unsharp-mask / Richardson-Lucy hybrid using that measurement as the kernel size. This produces tighter stars and reveals fine structural detail in nebulae.

Strength options:

Light — Subtle sharpening. Safest for any stack; excellent starting point.

Medium — Good balance.

Strong — Aggressive; best for high-SNR stacks with good seeing conditions.

Note: Sharpening amplifies both signal and noise. If your stack is already noisy, try using Denoise before Sharpen (the pipeline applies Sharpen first, then Denoise).

If Drizzle is also enabled, the FWHM estimate is automatically scaled up to account for the doubled pixel scale of the drizzled output.

7.7 DRIZZLE (2x RESOLUTION)

Drizzle is a technique that produces an output image at double the native pixel resolution of your sensor. Rather than simply scaling up (which adds no new information), drizzle exploits the tiny sub-pixel position differences between aligned frames to reconstruct detail beyond the native pixel scale.

How it works:

Each input pixel is projected onto a finer output grid with a slightly shrunken "drop" (the pixel footprint). Because each frame is shifted by a sub-pixel amount relative to the others (from natural tracking imperfections and dithering), different frames contribute flux to slightly different output pixels. Averaging over many such contributions reconstructs sub-pixel detail.

The output image is 4x larger in file size (2x in each dimension). This is normal and expected.

Works best when:

- Your mount or guiding software dithers between frames (recommended).

```
    Dithering introduces random sub-pixel offsets, giving the drizzle
```

```
    algorithm more independent samples per output pixel.
```

- You have 15 or more frames (more frames = better sub-pixel sampling).
- Your seeing is reasonable (drizzle cannot recover detail lost to bad seeing).

Important: When Drizzle is enabled alongside Auto Plate Solve, the app automatically halves the scale hints sent to nova.astrometry.net to account for the doubled resolution. This is handled transparently — you do not need to adjust your scale hints manually.

7.8 AUTO PLATE SOLVE AFTER STACKING

When enabled, the app automatically submits the completed stack to nova.astrometry.net for plate solving immediately after stacking finishes.

Requirements:

- A valid API key entered in the Plate Solve tab.
- Scale hints calculated from your focal length and pixel size (strongly

```
recommended — solving without scale hints can take 10+ minutes or fail).
```

If scale hints are not set when you tick this checkbox, the app will display a reminder.

After the auto-solve completes, the WCS astrometry data is automatically embedded into your stacked FITS file. The solve result (coordinates, field size, objects detected) is also displayed in the progress log.

If no API key is entered, auto plate solve is skipped and the pipeline completes with a note in the log.

See Section 11 for full details on plate solving.

8. RUNNING THE PIPELINE

8.1 STARTING A STACK

Click the "Start Processing" button in the Progress panel at the bottom right.

The button is a large orange-bordered ring labelled "Start Processing".

At least one light frame must be loaded before it is usable.

8.2 THE PROGRESS LOG

The text area in the Progress panel shows a real-time log of every step:

- Master frame building or loading
- Frame loading and parallel calibration
- Debayering (colour cameras only)
- Auto frame rejection scores (if enabled)
- Alignment progress
- Local normalisation (if enabled)
- Stacking method used
- Auto-crop dimensions (if enabled)
- Gradient removal (if enabled)
- Sharpening and/or denoising (if enabled)
- Output file path and dimensions
- Auto plate solve progress (if enabled)
- Final "Done!" message

The frame status bar at the very bottom of the window shows the total frame count, accepted frame count, and any rejected frame count after the run.

8.3 CANCELLING A RUN

While the pipeline is running, the Start Processing button changes to "Cancel". Click it to request cancellation. The pipeline checks for cancellation between each major stage, so it may take a moment to stop.

8.4 WHAT HAPPENS BEHIND THE SCENES

The pipeline executes these stages in order:

Stage 1: Build master calibration frames

- If you provided dark frames, they are median-stacked into a master dark,

```
        which is saved to your output folder as master_dark.fits.
```

- If you provided flat frames (and optionally dark flat frames), a master

```
        flat is built and saved as master_flat.fits.
```

- If you loaded pre-existing master frames, they are used directly.

Stage 2: Load and calibrate light frames (parallel)

- All light frames are loaded in parallel across available CPU cores.
- Each frame is dark-subtracted and flat-corrected simultaneously.
- If camera type is Colour, each frame is debayered from Bayer to RGB.

Stage 2b: Auto frame rejection (if enabled)

- Star profiles are measured for each calibrated frame.
- Poor-quality frames (blurry, trailed) are removed from the set.

Stage 2c: Local Normalisation (if enabled)

- Gradient removal is applied to each calibrated, debayered frame

```
        individually, before alignment.
```

- Runs at this stage so that all pixels are valid (no NaN alignment

```
        borders) and the colour-aware gradient model works correctly.
```

Stage 3: Align frames

- All frames are aligned to the chosen reference frame using star pattern

```
        matching (triangle similarity, via the astroalign library).
```

Stage 4: Stack

- All aligned frames are combined using the chosen stacking method.
- If Drizzle is enabled, drizzle stacking replaces the normal stack.

Stage 4b: Post-processing

- Auto-crop (if enabled)
- Gradient removal on the stack (if enabled)
- Sharpening (if enabled)
- Denoising (if enabled)

Stage 5: Save

- The result is saved to the output path as FITS (or XISF).
- A preview is shown in the Preview panel.

- The completion chime sounds.
- Auto plate solve runs (if enabled).

9. PREVIEW PANEL & HISTOGRAM

The Preview panel fills the top-right portion of the window and shows the currently selected file or the most recently stacked result.

The image is automatically stretched using an auto-stretch algorithm (midpoint histogram transfer) so faint detail is visible even in a raw, linear FITS file. This stretch is display-only — it does not affect the saved file, which is always in linear (unstretched) form.

Click any file in the Stacking tab's frame lists to preview it.

After a stack completes, the stacked result is automatically displayed.

The Histogram panel sits to the right of the Preview panel and shows the pixel value distribution of whatever is currently in the Preview. The X axis represents pixel brightness (left = dark, right = bright) and the Y axis represents how many pixels have each brightness value. For a well-exposed stack, the main peak should sit in the lower-left portion of the histogram with a tail extending toward brighter values.

10. POST-PROCESSING WINDOW

The Post-Processing Window opens any stacked image for interactive fine-tuning. It is separate from the pipeline processing options in the Settings panel — use it after stacking to apply finishing touches, then compare directly with the original before saving.

To open: after a stack completes, click File > Post-Process... or use the post-process button that appears in the preview toolbar. The window opens maximised to fill your screen and works on the currently displayed image.

Processing is cumulative — each Apply updates the working result, which becomes the starting point for the next Apply. This lets you build up adjustments step by step. Up to 5 Undo levels are available at any time.

Your original stacked file on disk is never modified.

The controls panel on the right is organised into six steps, shown in the indicator bar at the top of the panel:

❶ Colour Balance — correct green casts and channel imbalances
❷ Tone — adjust Brightness, Contrast, and Saturation
❸ Background — remove gradients and crop alignment borders
❹ Enhance — sharpen and denoise
❺ Stars — reduce dominant star brightness
❻ Save — save the result as FITS, TIFF, JPEG, or PNG

10.1 HOW THE PREVIEW WORKS

The window has a large image preview on the left and a controls panel on the right. Use the scroll wheel to zoom in and out on the preview, and click and drag to pan.

Stretch controls

A "Stretch" dropdown above the preview selects how the image is displayed. All stretching is display-only — the underlying data is always saved as linear float32.

```
Auto – Gentle   Subtle stretch. Background appears lighter. Good for
```

```
                judging faint structure without clipping highlights.
```

Auto – Normal	The default setting. Moderate background brightness
	suited to most targets.
Auto – Strong	More aggressive midtone pull. Useful for revealing faint
	emission or targets buried in a bright region.
Auto – Max	Maximum stretch. Very dark background. For extremely
	faint targets.
Linear	No stretch. Pixel values displayed as-is. Most images
	will appear nearly black in Linear mode — this is correct
	for a raw linear FITS file.

True-colour preview (what you see is what you get)

Raw Bayer-pattern images from colour cameras carry a strong green tint — the RGGB sensor pattern has two green photosites per four-pixel block, producing roughly twice the green signal compared with red or blue.

The MAIN window's quick-look preview hides this automatically with a display-only correction. The Post-Processing window deliberately does NOT: here the preview always shows the true colour state of the data, so the image will look green when it first opens. This is honest — it is what your data really looks like, and it is what would be saved.

To fix it for real, use Colour Balance ❶ (already ticked, with Auto mode on by default) and click Apply. Because the preview is truthful, every colour tool shows its genuine effect, and the saved JPEG/TIFF/PNG always matches the screen exactly.

Cumulative processing and Undo

Each time you click Apply, the processed result becomes the new working image. The next Apply starts from this updated state. This lets you chain adjustments: apply Colour Balance first, then Tone, then gradient removal — each step refines the result of the previous one.

The Undo button (■) reverses the last Apply. Up to 5 Undo levels are stored. Click Reset to Original to discard all processing and return to the original stack.

Show Original

Toggle this button at any time to compare the current processed state with the original stacked image. The preview switches between the two at the same stretch scale so brightness can be compared fairly.

10.2 COLOUR BALANCE (❶)

Adjusts the relative brightness of the red, green, and blue channels. Use this to correct a colour cast from:

- Light pollution (orange/green tint from city skies)
- Atmospheric airglow (reddish tint from oxygen and sodium emission)
- Bayer sensor channel imbalance (green excess from RGGB sensors)
- Narrowband filter leakage

Auto (recommended)

Samples the sky background from the image corners and automatically neutralises any colour cast. Both "Enable colour balance" and "Auto (recommended)" are already ticked when the window opens — just click Apply for a one-click colour correction.

If the target fills the whole frame (a large emission nebula or dense Milky Way field), the corners contain nebulosity rather than clean sky. Auto detects this automatically and switches to sampling the darkest 5 % of the whole image instead, so full-frame targets balance correctly too.

Manual adjustment

Uncheck "Auto (recommended)" to access the R, G, and B sliders.

Each channel has a multiplier from 0.50× to 2.00×:

- 1.00× = no change (default).
- Above 1.00× = brighter (boost the channel).
- Below 1.00× = dimmer (reduce the channel).

Each slider has a paired numeric spinbox — type a value directly (e.g. 1.35) for precise entry without dragging.

Colour balance has no effect on monochrome (single-channel) images.

Recommended workflow: run Colour Balance first (❶) before Tone (❷). The Auto mode neutralises the sky cast to a flat grey, giving Tone adjustments a clean baseline to work from.

10.3 TONE ADJUSTMENT (➋)

Adjusts the global brightness, contrast, and colour saturation of the image.

All three controls are applied together in a single Apply pass.

Brightness –100 … +100 %. Scales all pixel values exponentially

```
(2^(amount/100) factor). +100 doubles the brightness; –100
```

```
halves it. Each step feels perceptually equal, like
```

```
photographic exposure stops.
```

Contrast –100 … +100 %. Expands (positive) or compresses (negative)

```
the pixel value range relative to the sky floor. The sky
```

```
background stays anchored while bright nebulae and stars are
```

```
pulled further from it (positive) or pushed closer (negative).
```

Saturation –100 … +100 %. Scales colour deviation from grey. +100

```
doubles the vividness of all colours; –100 converts the
```

```
image to greyscale; 0 leaves colours unchanged. Only affects
```

```
colour (RGB) images — monochrome images are returned unchanged.
```

How to use:

1. Tick "Tone adjustment" to enable the section.

2. Set Brightness, Contrast, and Saturation using the sliders or

```
spinboxes.
```

3. Click Apply and use "Show Original" to compare.

Because processing is cumulative, a second Tone Apply adds on top of the first. Click Undo (■) to reverse if the result overshoots.

10.4 BACKGROUND (❸)

Auto-crop edges

Trims dark alignment borders from the stacked result. Same operation as Auto-Crop in the pipeline (Section 7.4), applied interactively here. Safe to enable if Auto-Crop was not used during the pipeline run.

Remove gradient

Fits and subtracts a smooth sky background surface to correct light pollution, moonlight, or vignetting. Same algorithm as Section 7.2 (Remove Light Pollution Gradient), applied to the stack interactively.

10.5 ENHANCE (❹)

Sharpen

PSF-informed sharpening. See Section 7.6 for full details.

Presets: Light / Medium / Strong.

Denoise

Non-Local Means denoising. See Section 7.5 for full details.

Presets: Light / Medium / Strong.

10.6 STAR BRIGHTNESS REDUCTION (❺)

Reduces the brightness of stars in the stacked image, making faint nebulosity and background structure more visible relative to dominant stars. Useful when bright stars overpower a widefield target, or when you want to reveal faint emission structure near saturated stars.

How it works:

1. A high-pass filter isolates point sources (stars) from the soft

```
    background (nebulosity, gradients). Stars are sharp and appear in the
```

```
    high-pass layer; nebulosity is blurry and is filtered out.
```

2. Peak detection finds individual star centres across the full image.
3. A soft Gaussian mask is painted over each detected star, sized to cover

```
    the star's disc and outer halo.
```

4. For each colour channel, the stellar signal above the local background

```
is estimated and partially subtracted, scaled by the Strength setting.
```

Background estimation uses two scales:

- Near (25-pixel Gaussian) — used away from bright stars.
- Far (100-pixel Gaussian) — used inside star masks. The near estimate is

```
contaminated by the star's own halo flux; the far estimate averages a
```

```
much wider area, giving a more accurate sky level and allowing reduction
```

```
to reach cleanly into the outer halo without leaving a glowing ring.
```

The sky, nebulosity, and dust structure beneath stars are preserved. Stars embedded deep inside bright nebula cores may be only partially reduced (the background estimator is elevated by surrounding emission), but the result is always physically plausible — no dark holes or halos are created.

No AI or model files are required. The algorithm runs entirely on the CPU using classical image processing (scipy.ndimage, scikit-image).

How to use:

1. Tick "Reduce stars" to enable.
2. Set Strength: drag the slider or type directly into the spinbox

```
(0 % = no change, 100 % = maximum reduction). Start at 50–70 %.
```

3. Click Apply and use "Show Original" to compare the result.
4. Adjust Strength and Apply again if needed.

10.7 CROP TOOL

Selects a rectangular region of the image to keep, discarding everything outside it. Useful for:

- Removing edge artifacts or alignment borders not caught by Auto-crop
- Tightly framing a specific target within a wide-field stack
- Excluding a very bright star or lens artifact near the frame edge

How to use:

1. Click "✂ Crop" in the toggle bar below the preview image.

```
The cursor changes to a crosshair.
```

2. Click and drag on the preview to draw a selection rectangle.

```
An orange rubber-band rectangle shows the crop region as you drag.
```

3. Release the mouse. The crop region is confirmed automatically. The

```
dimensions and position appear in the toggle bar
```

```
(e.g. "Crop: 3840×2160 px at 120,80").
```

4. Click Apply. The crop is applied to the current working result

```
(which may already include previously applied steps). To crop the
```

```
original stack, click Reset to Original first.
```

To remove the crop selection:

Click "✕ Clear" (appears once a crop is set), or click Reset to Original.

Notes:

- Crop mode exits automatically after you release the mouse.
- A minimum selection of 10×10 pixels is required to confirm a crop.
- The crop selection is remembered across Apply passes until cleared.

10.8 APPLYING CHANGES AND SAVING (⑥)

■ Apply

Runs all ticked options on the current working image and updates the preview. Processing runs on a background thread — the Apply button is greyed out until the run completes. A status message at the bottom of the controls panel shows progress.

After Apply completes, the result becomes the new working image. The next Apply starts from here, letting you chain adjustments step by step.

■ Undo

Reverses the last Apply, restoring the working image to its state before that Apply ran. Up to 5 Undo levels are stored.

Reset to Original

Discards all processing and returns the working image to the original unprocessed stack. All control settings are preserved so you can reconfigure and Apply again.

Save FITS...

Saves the current working result as a 32-bit floating-point FITS file — the best format for further processing in PixInsight, Siril, or any other tool. If no processing has been applied, the original stack is saved.

Save TIFF... / Save JPEG... / Save PNG...

Saves a stretched, display-ready version of the current result. Each format applies a fresh auto-stretch at save time, so the saved image is correctly exposed regardless of the current preview stretch mode.

TIFF — 8-bit stretched. Good for Photoshop, GIMP, or Lightroom.

JPEG — 8-bit compressed. Best for web sharing and online posting.

PNG — 8-bit lossless. Good for sharing without compression artefacts.

Close

Closes the window. Your original stacked FITS file on disk is never affected. Any processed result not explicitly saved is discarded.

11. PLATE SOLVE TAB

11.1 WHAT IS PLATE SOLVING?

Plate solving is the process of identifying the exact position of an astrophotograph in the sky by matching the pattern of stars in the image against a database of known star positions. The result tells you:

- The Right Ascension (RA) and Declination (Dec) at the centre of the image
- The field of view in arcminutes
- The pixel scale in arcseconds per pixel
- The image orientation (position angle, east of north)
- Any named deep-sky objects in the field
- Full WCS (World Coordinate System) data for embedding in the FITS file

WCS data embedded in a FITS file allows tools like PixInsight, Siril, ASTAP, and Aladin to overlay grids, identify objects, and perform spectrophotometric colour calibration (SPCC in PixInsight requires WCS).

Haysey's Astrostacker uses the free nova.astrometry.net service for plate solving, which means an internet connection is required for this feature.

11.2 GETTING A FREE API KEY

An API key is strongly recommended. Without it, solving goes into an anonymous queue which is much slower (can be 30+ minutes).

1. Open a browser and go to: https://nova.astrometry.net
2. Click "Sign in / Register" and create a free account.
3. Once logged in, go to: https://nova.astrometry.net/api_help
4. Your API key is shown on that page — it looks like: abcdefghijklmnop
5. Copy it and paste it into the API Key field in the Plate Solve tab.

```
Tick "Show key" if you want to verify it.
```

Your API key is saved automatically when you type it and persists across sessions. You only need to enter it once.

11.3 SETTING SCALE HINTS (STRONGLY RECOMMENDED)

Scale hints tell nova.astrometry.net the approximate pixel scale of your image (in arcseconds per pixel). Without them, the solver must search through every possible scale — this can take 10–30 minutes. With accurate scale hints, it typically solves in 30–90 seconds.

The app includes a Scale Hints calculator. Enter your focal length and camera pixel size and click "Calculate & Apply":

Focal length — your telescope's focal length in millimetres

Pixel size — your camera's pixel physical size in micrometres (µm)

The app calculates your pixel scale as:

arcsec/pixel = (pixel_size_µm ÷ focal_length_mm) × 206.265

Then applies a ±25% tolerance band (e.g. a result of 2.0 arcsec/px becomes a range of 1.5–2.5 arcsec/px) to give the solver a comfortable window.

These values are saved automatically when you click Calculate & Apply and are reloaded the next time you open the app. You only need to enter them once per telescope/camera combination.

You can also type scale bounds manually in the Lower bound and Upper bound fields in Arcsec/pixel, Arcmin (field width), or Degrees (field width) units.

11.4 FINDING YOUR FOCAL LENGTH

Your telescope focal length is usually printed on:

- The focuser drawtube or the OTA tube itself (e.g. "f=500mm")
- The product label on the front or rear of the scope
- Your telescope's specification sheet or product page

Common examples:

Celestron NexStar 8SE — 2032 mm

Sky-Watcher Esprit 100ED APO — 550 mm

William Optics RedCat 51 — 250 mm

Explore Scientific AR102 (f/6.5) — 663 mm

Sky-Watcher 200P (8" f/6) — 1200 mm

Rowe-Ackermann Schmidt Astrograph (RASA) 8" — 400 mm

If you use a focal reducer or Barlow lens, you must account for it:

Focal reducer ×0.7 at 1000mm scope → 1000 × 0.7 = 700 mm effective

Barlow ×2 at 600mm scope → 600 × 2 = 1200 mm effective

11.5 FINDING YOUR CAMERA PIXEL SIZE

Pixel size is a physical property of your camera sensor. It is constant for
a given camera model and does not change with settings. It is measured in
micrometres (µm, also written "microns").

Where to find it:

- The camera's specification sheet or product page on the manufacturer's

```
    website (look for "pixel size", "pixel pitch", or "pixel scale")
```

- Astronomy camera databases such as:

```
      https://www.cloudynights.com/   (search your camera model)
```

```
      https://astronomy-imaging-camera.com/  (ZWO cameras)
```

Common examples:

ZWO ASI294MC Pro — 4.63 µm

ZWO ASI2600MC Pro — 3.76 µm

ZWO ASI183MC Pro — 2.40 µm

ZWO ASI071MC Pro — 4.78 µm

Canon EOS R5 — 4.40 µm

Canon EOS 600D (T3i) — 4.30 µm

Nikon D7500 — 4.23 µm

Sony A7 IV — 5.95 µm

Sony A7S III — 9.44 µm

Altair 26C — 3.76 µm

Player One Neptune-C II — 2.90 µm

Note: If your camera supports hardware binning (2×2, etc.) and you capture
in binned mode, multiply the pixel size accordingly:

e.g. ZWO ASI294MC Pro at 2×2 binning: 4.63 × 2 = 9.26 µm effective

---

11.6 MANUAL PLATE SOLVING

To manually plate solve any image:

1. Select the image using the Browse... button in the Image section of the

```
     Plate Solve tab. (Alternatively, click a file in the Stacking tab —
```

```
     it is automatically set as the plate solve target.)
```

2. Ensure your API key is entered and scale hints are set (see 11.3).

3. Click the "Solve" button.

The progress log below the button shows each step:

- Connecting to Astrometry.net
- Uploading the image (file size shown in MB)
- Waiting for a job to be assigned
- Solving status with elapsed time
- Result retrieval (calibration data, object list, annotations, WCS)

Note: The app uploads a version of your image stretched by 20% in each dimension. This improves the readability of the annotated result returned by the server. The WCS data is automatically corrected back to the original image coordinates before being saved.

While a solve is running, the Solve button becomes a red "Cancel" button. Click it to abort.

11.7 THE SOLVE RESULT WINDOW

After a successful solve, a result window opens automatically showing:

- The image with named objects (NGC/IC catalogues, named stars) circled

```
      and labelled with their common names
```

- A text summary of the solution in the Results box:

```
        RA, Dec, orientation, field size, pixel scale, parity, and object list
```

If you close the result window, you can regenerate the result by solving again.

11.8 EMBEDDING WCS INTO ANY FITS FILE

After a successful solve, the "Write WCS to FITS..." button becomes active.

Click it to choose any FITS file and embed the astrometry WCS keywords into it.

This is useful when you want to plate-solve one version of an image (e.g. a quick export) and then embed the WCS into a higher-quality stack of the same field.

The following WCS keywords are embedded:

CTYPE1, CTYPE2 — coordinate axis types (RA---TAN, DEC--TAN)

EQUINOX, RADESYS — J2000 / ICRS reference frame

CRVAL1, CRVAL2 — RA/Dec of the reference pixel

CRPIX1, CRPIX2 — reference pixel coordinates

CD1_1 to CD2_2 — CD matrix (rotation + scale in degrees/pixel)

OBJCTRA, OBJCTDEC — human-readable RA/Dec (read by PixInsight)

PIXSCALE, PA — pixel scale and position angle

PLTSOLVD — flag indicating a plate solve was performed

FLDWIDTH, FLDHGHT — field size in arcminutes

11.9 AUTO PLATE SOLVE AFTER STACKING

See Section 7.8. WCS data from the auto-solve is embedded into the stacked FITS file automatically. You do not need to click "Write WCS to FITS..." again.

12. MOSAIC TAB

The Mosaic tab allows you to stitch multiple separately stacked and plate-solved panels into a single wide-field image.

Requirements:

- Each panel must be a FITS file with WCS astrometry embedded.

```
    (Use the Plate Solve tab to plate-solve each panel first.)
```

- At least 2 panels.

How to use:

1. Click "Add Panels" and select all the plate-solved FITS files.
2. Set the output path using Browse... (defaults to "mosaic.fits").
3. Click "Build Mosaic".

The mosaic engine uses the WCS coordinates in each file to determine where each panel sits in the sky, then reprojects and blends them into a common output grid. The output appears in the Preview panel when complete.

Tips:

- Panels should overlap by 10–20% for best blending.
- Use the same camera, scope, and settings for all panels for consistent

```
    brightness and colour matching.
```

- All panels must be plate-solved with the same or compatible WCS reference

```
    frame (J2000/ICRS, which is what nova.astrometry.net provides).
```

13. TOOLS MENU

13.1 BLINK COMPARATOR

Tools > Blink Comparator...

Opens a window that cycles rapidly through all loaded light frames, allowing you to visually compare them for quality, trailing, tracking errors, or atmospheric turbulence. This is a quick way to identify frames you want to remove before stacking.

Controls:

< Prev / Next > — Step through frames manually.

Play / Stop — Automatically cycle through all frames.

Speed slider — Set the blink interval from 100ms to 2000ms.

```
Drag left for faster blinking, right for slower.
```

Each frame is auto-stretched for display so all frames appear at consistent brightness regardless of their actual pixel values.

Requires at least 2 light frames to be loaded.

13.2 VIEW FITS HEADER

Tools > View FITS Header...

Opens the FITS header viewer showing all metadata keywords stored in a FITS file. Tries the currently selected plate solve image first, then the stacked output path, then asks you to pick a file.

Useful for checking exposure metadata (EXPTIME, DATE-OBS, TELESCOP, etc.), verifying that WCS keywords were embedded correctly, and inspecting camera settings recorded during capture.

13.3 SETUP WIZARD

Tools > Setup Wizard...

Re-opens the first-run Setup Wizard at any time. Useful for updating your camera type, API key, focal length, or pixel size. Any values you change are saved when you click through to the end of the wizard.

13.4 ABOUT

Tools > About Haysey's Astrostacker...

Displays the app version, build codename, acknowledgements, and a brief summary of features. Also shows where to find the GETTING_STARTED.txt beginner guide included with the app.

14. FILE MENU

14.1 SAVE SESSION / LOAD SESSION

File > Save Session...

File > Load Session...

Sessions are JSON files that store all current settings and file paths in one place, so you can pick up exactly where you left off, or quickly switch between different imaging projects.

What is saved:

- All loaded file paths (lights, darks, flats, dark flats, master frames)
- Camera type and Bayer pattern
- Stacking method and all parameters
- Output file path
- Reference frame number
- All Processing checkbox states (auto-reject, gradient, local norm, etc.)
- Denoise and sharpen strengths
- Auto plate solve state

Note: Session files store file paths, not the image data. If you move or delete the original files, the paths in the session will no longer be valid. Files that no longer exist at the stored path are silently skipped when loading.

14.2 EXPORT AS TIFF / PNG

File > Export as TIFF...

File > Export as PNG...

Exports the currently displayed preview image (the stacked result or the last manually previewed frame) as a TIFF or PNG file with automatic stretching applied.

TIFF — 16-bit stretched image. Preserves more tonal range than PNG. Suitable

```
            for further processing in Photoshop, GIMP, or Lightroom.
```

PNG — 8-bit stretched image. Smaller file size. Good for web sharing.

The export stretch is calculated from the current preview's data. For best results, export immediately after stacking.

Note: These exports are for sharing and visual inspection. For further astrophotography processing, always use the FITS output file — it contains the full linear, unprocessed data.

15. STACKING STRATEGY GUIDE

15.1 CHOOSING THE RIGHT STACKING METHOD

Quick reference:

Frames available Best method(s)

[redacted]

Any number Median (safe default — use this if unsure)

3–10 frames Percentile Clipping or Winsorized Sigma

10–20 frames Winsorized Sigma or Percentile Clipping

20+ frames Sigma Clipping (best SNR with good rejection)

30+ frames Sigma Clipping or Weighted Mean (if seeing varied)

Varying quality Weighted Mean (Quality) or Noise-Weighted Mean

Very clean data Mean (Average) — maximum SNR, no rejection needed

If satellite trails or planes are a concern, Sigma Clipping and Percentile Clipping are both very effective at removing them. Median also works but leaves faint ghost trails at low frame counts.

15.2 HOW MANY FRAMES DO I NEED?

Stacking SNR improves with the square root of the number of frames. To halve your noise, you need 4× as many frames.

Frames SNR improvement over 1 frame

[redacted]

```
     2   1.4x
     5   2.2x
    10   3.2x
    20   4.5x
```

```
      50   7.1x
```

```
     100  10.0x
```

Practical minimums:

- Sigma Clipping: needs at least 10 frames for meaningful rejection

```
    (15+ recommended for best results).
```

- Percentile Clipping: effective from 5 frames.
- Auto-reject: requires at least 3 frames.
- Drizzle: benefits from 15+ frames.

15.3 CALIBRATION FRAME GUIDE

Frame type	Minimum	Recommended	Purpose
Darks	0	15–20	Remove thermal noise, hot pixels
Flats	0	15–20	Remove vignetting, dust donuts
Dark Flats	0	10–15	Calibrate flats more accurately
Bias	N/A	N/A	(not currently a separate step — use Dark Flats if needed)

For modern cooled astronomy cameras at -10°C or colder, darks become much less important. For uncooled DSLRs at ambient temperature, darks make a significant difference.

Flats are always beneficial if your images show vignetting (darkening toward corners) or dust donuts. They require matching the focus position of your lights exactly.

15.4 COLOUR VS. MONO CAMERAS

One-shot colour cameras (OSC / Bayer matrix cameras) have a colour filter array bonded to the sensor. Each photosite records only one of Red, Green, or Blue light. The app uses a process called debayering to reconstruct the full colour image from this mosaic pattern.

Pros of OSC: simpler workflow, faster imaging, one camera for all targets.

Cons of OSC: less efficient use of each photosite (each misses 2/3 of the

spectrum), colour-sensitive targets like emission nebulae show less Ha signal.
Monochrome cameras have no filter array and record all light with every photosite. They are used with separate colour filters (LRGB for broadband, or narrowband Ha/OIII/SII for emission nebulae).
For anyone just starting out, a one-shot colour DSLR or astronomy camera is the best choice. Set the app to Colour (Bayer) mode.

15.5 DRIZZLE — WHEN TO USE IT

Drizzle is most effective when:

1. You are imaging a high-detail target (galaxies, planetary nebulae, globular

```
clusters) where fine structure is important.
```

2. Your imaging system is "under-sampled" — the apparent size of your star

```
PSF (spread across the atmosphere and optics) spans fewer than 2 pixels.
```

```
Under-sampling means each seeing disc spreads across fewer than 2 pixels.
```

```
A rough guide: if your pixel scale is greater than half the seeing disc,
```

```
you are under-sampled. (e.g. 1.5 arcsec seeing at 1.0 arcsec/px = good
```

```
candidate for drizzle; 1.5 arcsec seeing at 0.3 arcsec/px = no benefit.)
```

3. You have 15 or more frames and dithering is enabled in your capture

```
software (N.I.N.A., SGP, APT, etc. all support dithering).
```

Drizzle is NOT recommended when:

- You have fewer than 10 frames.
- Your capture software does not dither between frames.
- Your pixel scale is already very fine relative to your seeing conditions

```
(over-sampled setup — drizzle would produce an artificially sharp but
```

```
noise-amplified result).
```

- You are time-limited and plan to stack more frames later.

16. TROUBLESHOOTING

■ The app doesn't start or crashes immediately.
On macOS, right-click the app and choose "Open" to bypass Gatekeeper. On Windows, allow it through SmartScreen. On Linux, ensure the AppImage has execute permission (chmod +x).

■ My stack is very noisy.
Stack more frames. Noise reduces with the square root of frame count. Also try enabling Denoise (Medium strength is a good starting point). Ensure you are using dark frames to remove hot pixel noise.

■ Stars in my stack are elongated or trailed.
Enable Auto-Reject to remove the worst frames before stacking. Check that your reference frame (Reference Frame spinner) is a sharp, well-tracked frame. Sigma Clipping with a low sigma value (2.0–2.5) may also help by rejecting the trailed pixels statistically.

■ Colours look wrong after stacking (e.g. nebulae are blue instead of red).
Check the Bayer Pattern setting. For most cameras, RGGB is correct. Try BGGR if colours are swapped. Consult your camera's spec sheet.

■ The stacked image has black triangle borders.
Enable Auto-Crop Stacking Edges to remove them automatically.

■ Plate solve fails with "no solution found".
- Check that your scale hints are set and correct (see 11.3). - Ensure your image actually contains detectable stars. - If using Drizzle, the app automatically halves the scale hints — if you are solving a drizzled image manually via the Plate Solve tab, halve your scale bounds manually before clicking Solve. - Try without scale hints first as a diagnostic (the solve will be much slower but confirms whether hints are the issue). - Check that nova.astrometry.net is accessible from your network.

■ Plate solving takes more than 15 minutes.
Set scale hints (see 11.3). Solving without hints is dramatically slower. Also check your API key is entered — anonymous users wait in a longer queue.

■ "Auto plate solve skipped — no API key set" message in the log.
Enter your API key in the Plate Solve tab (see 11.2). The key is saved across sessions — you only need to enter it once.

■ The pipeline fails with an alignment error.
Astroalign needs detectable stars in all frames. Check that: - Your frames are in focus and not massively over/underexposed. - You have at least a few dozen stars in the frame. - All frames are of the same target (different targets cannot be aligned together). For very crowded or very sparse star fields, try a different reference frame (change the Reference Frame spinner).

■ The pipeline runs but the output file is blank / all black.
This can happen if the stacking method produces no valid pixels (e.g. Sigma Clipping with very aggressive settings on only 3 frames). Try Median instead, then increase frame count.

■ I get an error about the master dark or flat being a different size.
This is normal and the app handles it automatically by resizing the calibration frame. The log will show a message like "Master dark is 4656×3520 but lights are 2328×1760 — resizing to match". No action required.

■ Mosaic doesn't stitch correctly or panels are in the wrong position.
Each panel must have WCS data embedded. Plate-solve every panel before attempting to build the mosaic. Ensure all panels overlap and were taken with compatible telescope/camera setups.

17. GLOSSARY

Arcsecond / Arcminute / Degree

Angular measurements of sky area. 1 degree = 60 arcminutes = 3600 arcseconds. The Moon is approximately 30 arcminutes (0.5 degrees) wide.

Bayer Pattern (RGGB, BGGR, etc.)

The arrangement of red, green, and blue photosites on a colour camera sensor. The most common is RGGB. Must match your camera's actual sensor layout for correct colour reproduction.

Calibration Frames

Auxiliary images used to correct systematic errors in light frames. Dark frames correct thermal noise. Flat frames correct vignetting and dust.

Dark Current

Electrons generated by heat in the camera sensor, independent of light. Increases with temperature and exposure duration. Subtracted by dark frames.

Debayer

The process of reconstructing a full-colour image from the raw Bayer mosaic pattern produced by a one-shot colour (OSC) camera.

Declination (Dec)

One of the two coordinates in the equatorial coordinate system. Declination is analogous to geographic latitude, measured in degrees from -90° to +90°. 0° is the celestial equator; +90° is the north celestial pole.

Drizzle

A stacking technique that reconstructs an output image at double the native pixel resolution by using sub-pixel position differences between aligned frames. Based on Fruchter & Hook 2002.

FITS (Flexible Image Transport System)

The standard file format in astronomy. Stores image data as floating-point arrays along with a text header of metadata keywords. The app reads and writes FITS files.

FWHM (Full Width at Half Maximum)

A measure of star size — the width of a star profile at half its peak intensity. Lower FWHM = sharper stars. Measured in pixels.

Hot Pixel

A malfunctioning photosite that always reads a very bright value regardless of the actual illumination. Dark subtraction removes most hot pixels.

Light Pollution

Artificial sky brightness from city lights, street lamps, etc. Produces an uneven background gradient in astrophotos. Corrected by gradient removal.

Linear / Stretched

A linear image has pixel values proportional to the actual amount of light collected. A stretched image has had its tonal range redistributed (usually making faint features more visible). The app saves linear FITS files and displays auto-stretched previews.

Master Dark / Master Flat

A stacked combination of multiple individual dark or flat frames. Stacking multiple frames reduces the noise in the calibration frame itself.

NLM (Non-Local Means)

A denoising algorithm that estimates each pixel's true value by averaging similar patches found elsewhere in the image. Very effective at preserving fine structure while reducing background noise.

Plate Solving

Identifying the precise sky position of an image by matching its star pattern against a reference catalogue. Produces RA, Dec, pixel scale, and WCS data.

PSF (Point Spread Function)

The shape that a perfect point source (like a distant star) produces in an image. Characterised by its FWHM and eccentricity. Used for sharpening and frame quality assessment.

Right Ascension (RA)

One of the two coordinates in the equatorial coordinate system. Analogous to geographic longitude, but measured in hours (0h to 24h) or degrees (0°–360°).

Seeing

The blurring of star images caused by atmospheric turbulence. Measured in arcseconds — lower is better. Typical values: 1–4 arcseconds.

Sigma

The standard deviation of a set of values — a measure of how spread out the values are. In sigma clipping, pixels more than N sigma from the median are rejected as outliers.

SNR (Signal-to-Noise Ratio)

The ratio of genuine signal (light from your target) to unwanted noise. Higher SNR = cleaner, more detailed image. Stacking N frames improves SNR by a factor of √N compared to a single frame.

Vignetting

Darkening toward the edges and corners of an image caused by the optics or filters. Corrected by flat frame calibration.

WCS (World Coordinate System)

A standard set of FITS header keywords that describe the mapping from pixel coordinates to sky coordinates. Allows tools like PixInsight and Siril to overlay grids and perform advanced analysis on plate-solved images.

XISF (Extensible Image Serialisation Format)

The native file format of PixInsight. Supported by the app for both reading and writing.

END OF USER MANUAL

Haysey's Astrostacker v1.0.0 Beta Bronze

17 sections · https://github.com/haysey/astrostacker